



TEEGALA KRISHNA REDDY ENGINEERING COLLEGE

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FACE RECOGNITION ATTENDANCE SYSTEM

INTERNAL GUIDE

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ABSTRACT

Automated face detection attendance systems leverage advanced computer vision and machine learning techniques to streamline and enhance traditional attendance tracking methods. By utilizing facial recognition algorithms, OpenCV, or deep learning models like Convolutional Neural Networks (CNNs), these systems can accurately identify and verify individuals in real time, eliminating manual entry errors and reducing administrative workload. The proposed system captures live video feed or images, detects faces, matches them against a pre-registered database, and records attendance automatically. Key benefits include improved efficiency, reduced proxy attendance, and seamless integration with existing databases. Experimental results demonstrate high accuracy and robustness under varying lighting and pose conditions, making it a reliable solution for educational institutions, corporate environments, and large-scale events.

EXISTING SYSTEM

The existing system for attendance management in most educational institutions is manual, where students sign an attendance sheet or use a fingerprint scanner. Here are some of the limitations of the existing system

DRAWBACKS:

- Time Consuming
- Inaccurate
- Buddy Punching
- Lack of Automation

PROPOSED SYSTEM

The proposed Face Recognition Based Attendance System aims to overcome the limitations of the existing system by using face recognition technology to mark attendance. Here are some of the key features of the proposed system

1. **Face recognition technology**
2. **Automated attendance marking**
3. **Improved accuracy**
4. **Elimination of buddy punching**
5. **Real-time attendance tracking**
6. **User-friendly interface**

ADVANTAGES:

- Increased Accuracy
- Improved Efficiency
- Enhanced security
- Reduce Administrative Burden



SYSTEM REQUIREMENTS

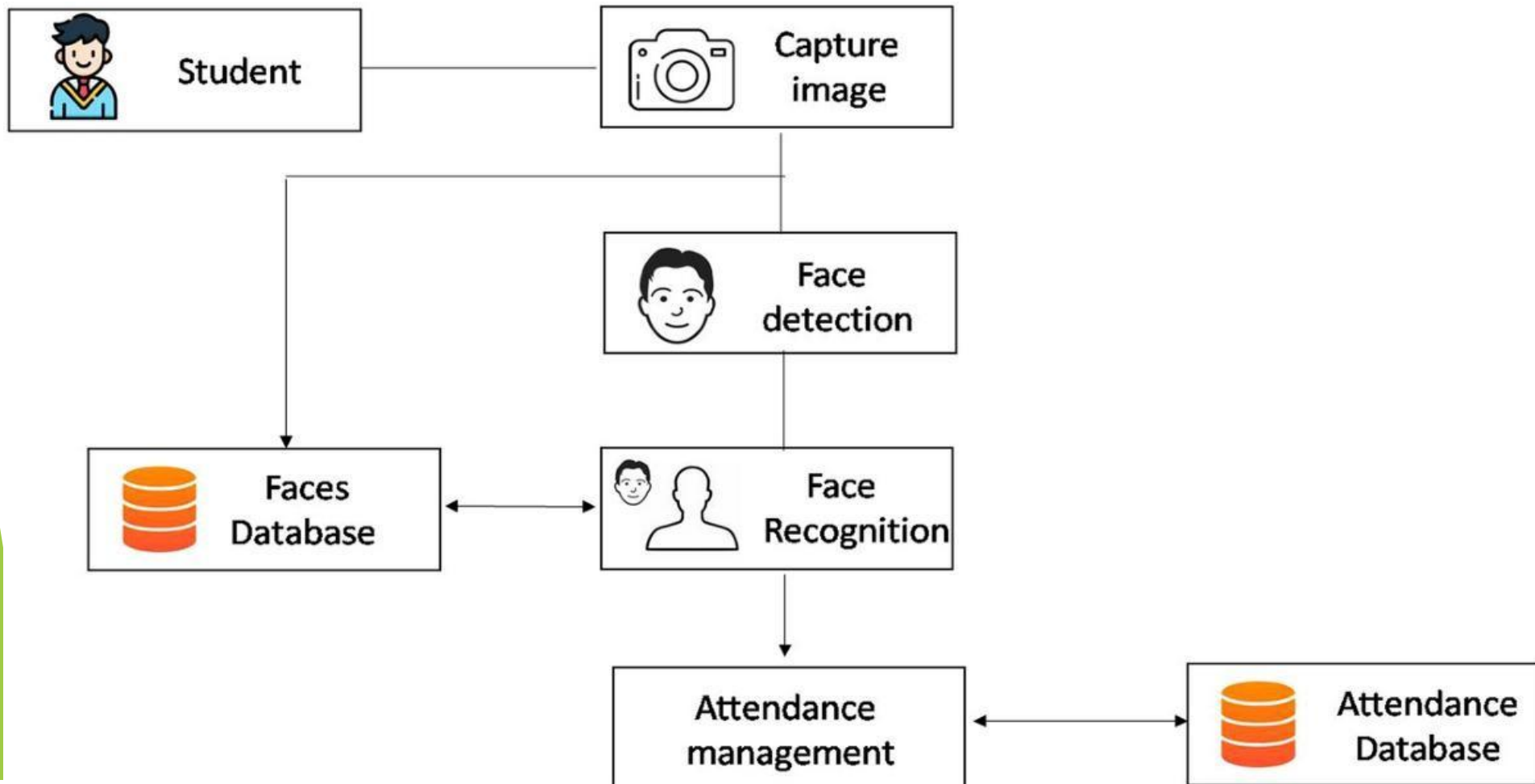
HARDWARE:

RAM	:	8 GB OR Higher
Storage	:	SSD (At least 128 GB)
GPU(optional)	:	NVIDIA GPU
Camera	:	HD webcam (720p or 1080p)

SOFTWARE:

Operating System	:	Window
Development Tools	:	Visual Studio

SYSTEM ARCHITECTURE



MODULES:

- Face Detection Module
- Face Recognition Modules
- Attendance Tracking Modules
- GUI Modules



Face Detection

Open CV

OpenCV's Haar cascade classifier is a pre-trained classifier used for face detection. It uses a cascade of classifiers to detect faces in images.

Face Recognition

Face recognition is the core component of face recognition-based attendance systems. The following module is used for face recognition:

Face Recognition Library:

Face recognition libraries such as `face_recognition` or `dlib` are used for face recognition. These libraries provide pre-trained models for face recognition and can be used to identify individuals.

Attendance Tracking

Attendance tracking is a critical component of face recognition-based attendance systems. The following module is used for attendance tracking:

Pandas:

Pandas is a library used for data manipulation and analysis. It is used to store and manage attendance records.

GUI Module

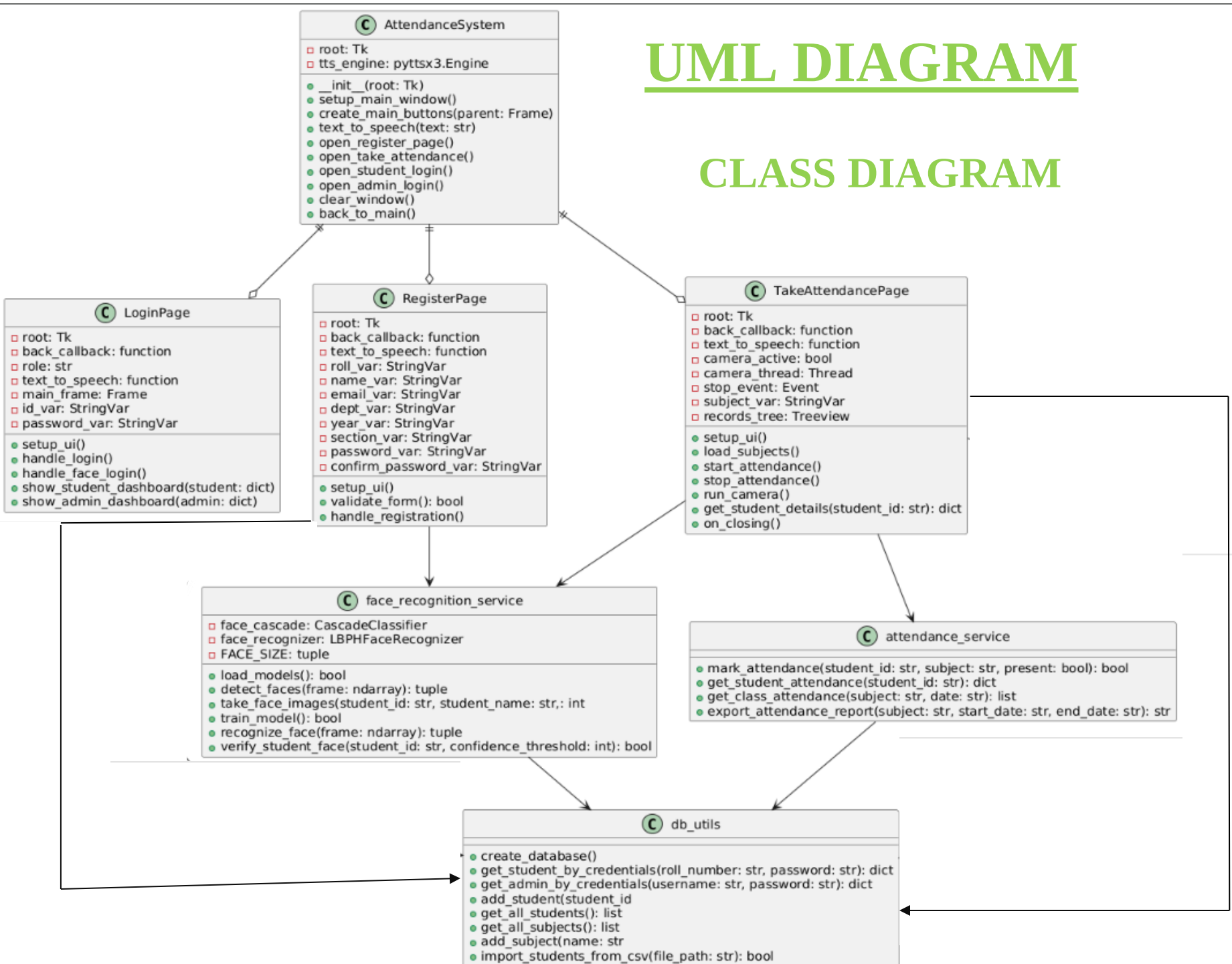
GUI modules are used to create a user-friendly interface for users to interact with the attendance system. The following module is used for GUI

Tkinter:

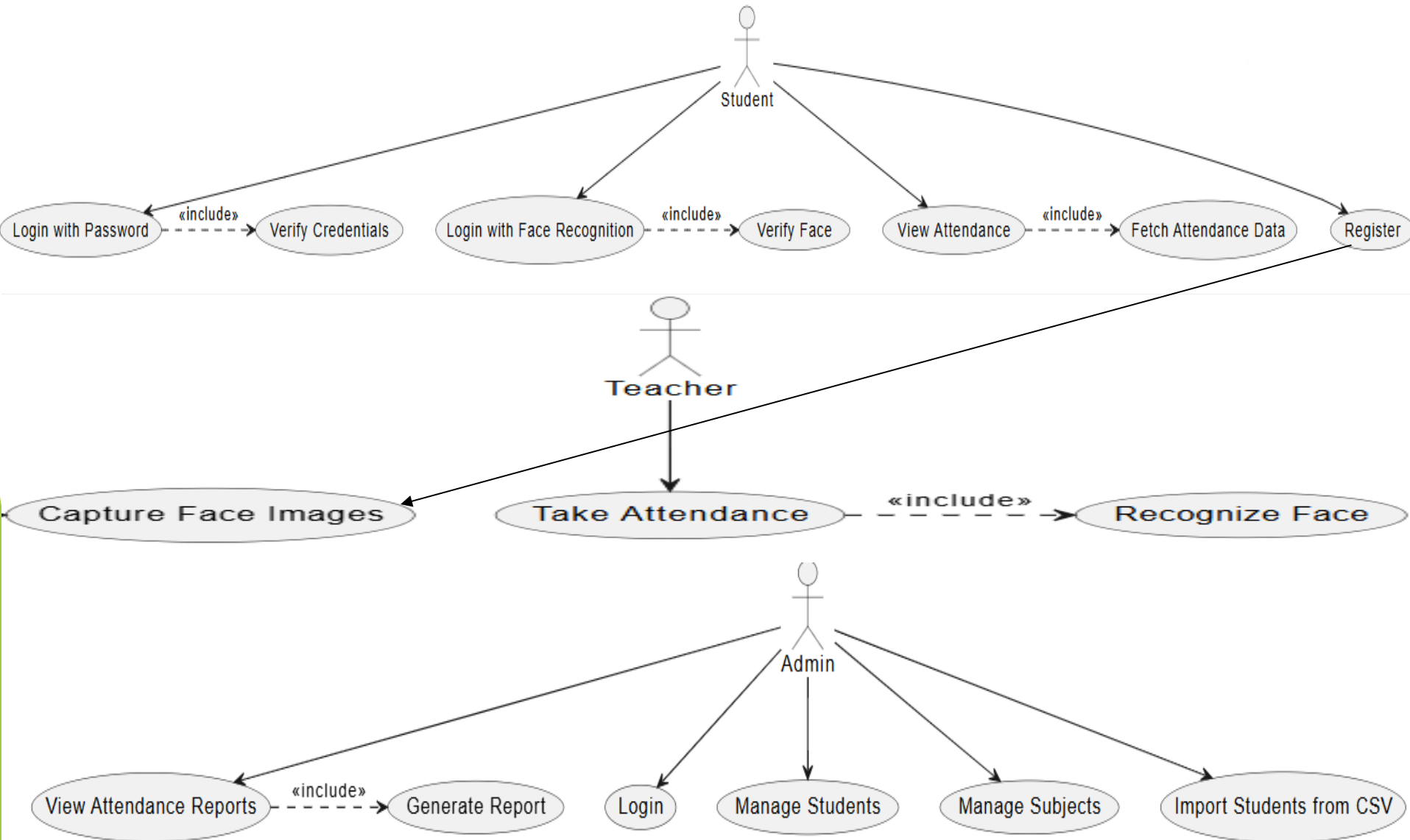
Tkinter is a library used for creating graphical user interfaces (GUIs). It provides a simple and easy-to-use interface for users to interact with the attendance system.

UML DIAGRAM

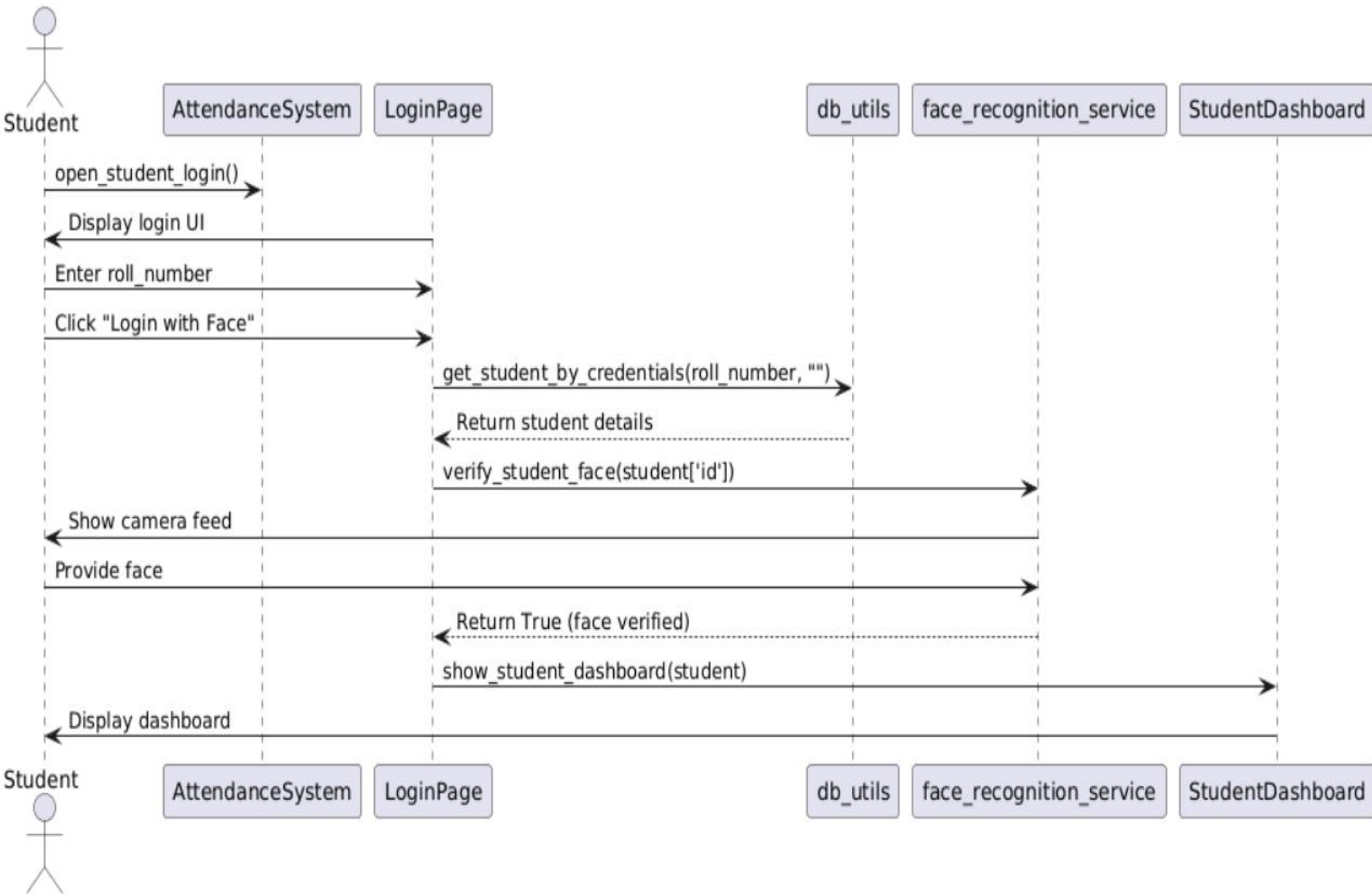
CLASS DIAGRAM



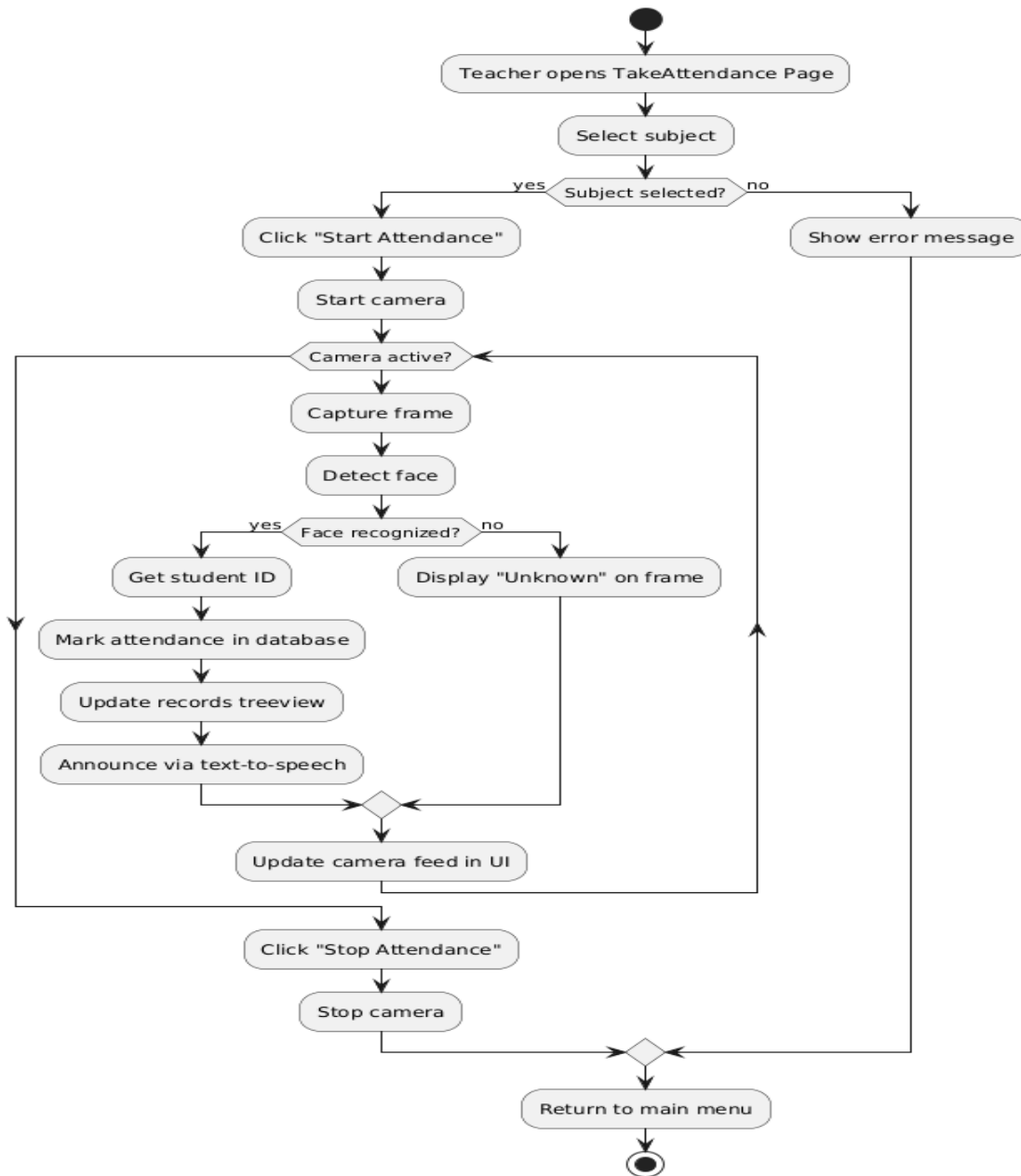
USE CASE DIAGRAM:



SEQUENCE DIAGRAM:



ACTIVITY DIAGRAM:



OUTPUT SCREENS:

TEEGALA KRISHNA REDDY ENGINEERING COLLEGE

Face Recognition Attendance System

Register New Student

Take Attendance

Student Login

Admin Login

Student Registration

Roll Number:	55
Full Name:	SRINU CHAWHAN
Email:	srinu@gmail.com
Department:	CSE
Year:	3rd Year
Section:	D
Password:	****
Confirm Password:	****

Register & Capture Face

Back

Take Attendance - Face Recognition Attendance System

Take Attendance with Face Recognition

Setup

Select Subject:

AI

Date:

2025-05-01

Start Attendance

Stop Attendance

Camera Feed

Attendance Records

Roll Number	Name	Time	Status
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Back to Home

Setup

Select Subject: AI

Date: 2025-05-01

Start Attendance

Stop Attendance

Al

2025-05-01

Status

Student Login

Roll Number:

Password:

Login

- OR -

Login with Face

Welcome, seenu

Roll Number: 55

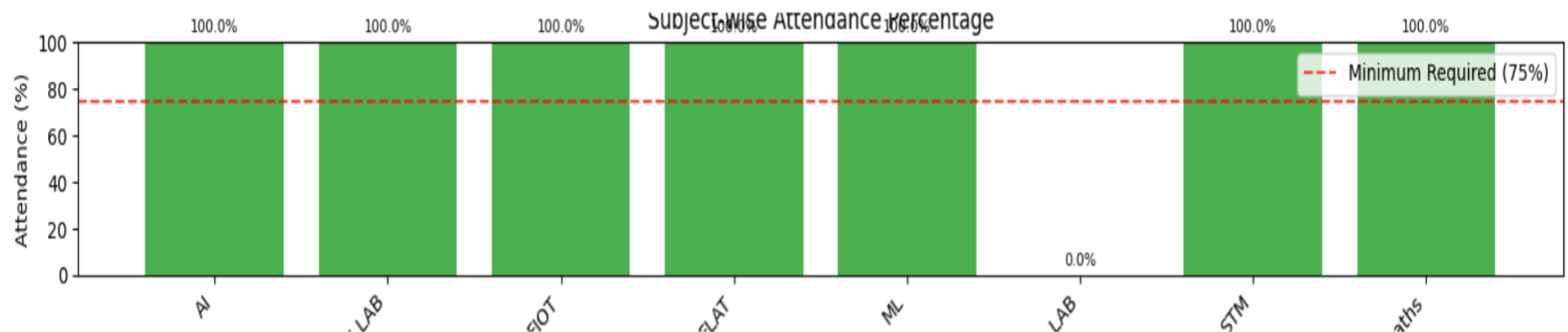
Attendance Summary

Overall Attendance: 87.5% Good standing

Subject-wise Attendance

Subject	Total Classes	Classes Attended	Attendance %
AI	1	1	100.0%
AI LAB	1	1	100.0%
FIOT	1	1	100.0%
FLAT	1	1	100.0%
ML	1	1	100.0%
ML LAB	1	0	0.0%
STM	1	1	100.0%
maths	1	1	100.0%

Attendance Visualization



Student Management

Add New Student

Import Students (CSV)

Search:

Search

Clear

Roll Number	Name	Department	Year	Section	Face Registered
-	-	-	-	-	No
123	DIVYA CH	CSE	3rd Year	D	Yes
22R91A05L6	SREENU CHAWHAN	CSE	3rd Year	D	Yes
55	seenu	cse	3rd Year	D	Yes
56	prabhas	cse	3rd Year	D	Yes
L3	RAMESH RATHOD	CSE	3rd Year	D	Yes
N1	MANASA	CSE	3rd Year	D	No
N2	S.MANASA	CSE	3rd Year	D	Yes

Logout

Subject Management

Add New Subject

Subject Name: Department:

Year: Semester:

Add Subject

All Subjects

ID	Subject Name	Department	Year	Semester	Created Date
1	AI	-	-	-	2025-04-29
6	AI LAB	-	-	-	2025-04-29
7	FIOT	D	3rd Year	2nd Semester	2025-04-29
2	FLAT	-	-	-	2025-04-29
5	ML	-	-	-	2025-04-29
8	ML LAB	CSE	3rd Year	2nd Semester	2025-04-30
4	STM	-	-	-	2025-04-29
3	maths	-	-	-	2025-04-29

Attendance Reports

Generate Report

Subject: Start Date:

End Date: (YYYY-MM-DD)

Generate CSV Report

Attendance Overview

CONCLUSION:

The system successfully achieves its objectives by enabling seamless student registration, real-time face recognition-based attendance marking, and comprehensive attendance reporting for students and administrators. Features like visual attendance analytics, CSV import/export, and audio feedback enhance usability and accessibility, catering to diverse user needs. While optimized for single-machine deployment, the system demonstrates reliability and efficiency in handling core functionalities, reducing manual effort and minimizing errors associated with traditional attendance methods.

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- [15] [14] Attendance System based on Face Recognition Venkata Kalyan Polamarasetty, Muralidhar Reddy Reddem,Dheeraj Ravi,Mahith Sai Madala
- Teegala krishna Reddy Engineering college attendance portal <http://tkrec.in/>

THANK YOU